

Process Characterization Report

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCR-004

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—

Role	Function	Name (synthetic)	Signature	Date
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AMV, analytical method validation; BLA, Biologics License Application; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; ICH, International Council for Harmonisation; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; NTU, nephelometric turbidity unit; PAR, proven acceptable range; PD, Process Development; qPCR, quantitative polymerase chain reaction; rcf, relative centrifugal field; RSD, relative standard deviation; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter.

Executive summary

Harvest and clarification recovers A-Mab from the production bioreactor and delivers a clarified, filtered feed to Protein A capture. Cells and cell debris are separated from the culture fluid by continuous disk-stack centrifugation. The fine debris and colloidal material the centrifuge passes are then removed on depth filters, and the clarified harvest is passed through a sterilising-grade filter before capture. The step forms no quality attribute of the antibody and no clearance of any impurity is claimed for it. What the step decides is the condition of the load presented to Protein A chromatography.

This report presents the characterization of the step on a qualified scale-down model, executed against the plan PCP-004. The scope came from the pre-characterization risk assessment RA-001, which found no parameter of this step with a credible route to a critical quality attribute and assigned all of them to univariate assessment. The step has 3 process parameters and all of them were carried into the study. Centrifugal field was studied between 6,000 and 12,000 g, depth filter load between 80 and 160 L/m², and post-clarification turbidity between 1 and 20 NTU. No designed multivariate experiment was performed and none is claimed.

At the set-point the step recovered 97.7% of the product mass entering it, which is 64,071 g of 65,596 g, and delivered a clarified harvest at 5 NTU. No quality attribute of the antibody changed measurably across the step at any setting studied. Host cell protein and residual DNA in the clarified harvest were the burden the culture delivered, and the harvest operation neither added to that burden nor cleared it within the precision of the assays. Raising the centrifugal field removes more debris and shears more cells, and only the first of the two appeared. Centrate turbidity fell as the field rose, and host cell protein in the clarified harvest did not rise at any field up to 12,000 g.

Of the 3 parameters, 2 were classified as key process parameters and 1 as a general process parameter. No parameter of this step is critical. Neither the centrifugal field nor the depth filter load changed host cell protein, residual DNA or aggregate in the clarified harvest at any setting studied. The normal operating ranges are unchanged from the ranges used for clinical manufacture and each sits inside the range characterized here.

During execution 2 deviations were recorded. One was a pre-use filter flush turbidity above its alert level, and one was a post-clarification turbidity excursion above the normal operating range on a single run at high depth filter loading. Both were investigated, both were dispositioned as retained, and neither affected product quality. The second is the clearest single piece of evidence in this study for the mechanism the step is controlled on.

The step contributes to the control strategy through two key process parameter ranges, in-process turbidity monitoring on every batch, depth filter media lot qualification, and the turbidity condition the clarified harvest must meet before it is loaded onto Protein A. This report rolls up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in a fed-batch mammalian cell culture and purified on a platform train that Novacyte Biologics has operated for three earlier products (CMC Biotech Working Group 2009). The drug substance process runs from the production bioreactor through harvest, Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration, and Figure 1.1 shows the train with the principal quality impact of each step. The commercial process operates at 15,000 L working volume, and the characterization reported here was performed on a scale-down model of that operation.

Harvest and clarification is Step 4 of that train. The culture fluid leaving the production bioreactor is fed continuously to a disk-stack centrifuge, which sediments cells and the larger cell debris under a centrifugal field. The centrate still carries fine debris, colloids and nucleic acid fragments that are too small to sediment at an industrial feed rate. That material is removed on a train of positively charged depth filters, and the clarified harvest is passed through a sterilising-grade filter into the capture step. The step is operated under SOP-2005 and SOP-2006.

The step forms no quality attribute of the antibody. The glycan, charge-variant and aggregate attributes are formed in the production bioreactor and are reported in PCR-003. Host cell protein and DNA clearance is credited to Protein A capture, cation exchange and anion exchange, and is reported in PCR-005, PCR-007 and PCR-008. Viral clearance is claimed for the low-pH hold, anion exchange and virus filtration, and no viral clearance is claimed for harvest. This step is therefore characterized for what it delivers to the next step and for its recovery of product, not for an attribute it sets.

A-Mab drug-substance process train

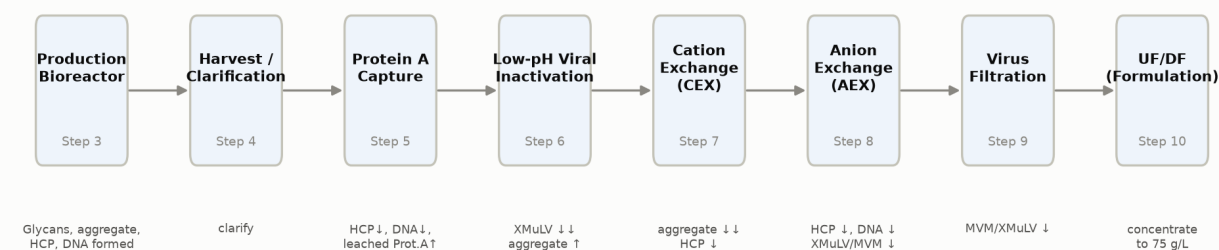


Figure 1.1: The A-Mab drug-substance process train, with the principal quality impact of each unit operation. Harvest and clarification is the primary recovery step between the production bioreactor and Protein A capture.

1.2 Objectives

This study had five objectives.

1. Quantify the recovery of product across the step at its set-point and bound the variability of that recovery between batches.
2. Establish the range of each process parameter over which the clarified harvest remains acceptable as a feed to Protein A capture.
3. Confirm that no quality attribute of the antibody is formed, modified or cleared across the step within the ranges studied.
4. Classify each parameter of the step and give the reason for the class.
5. Provide the parameter ranges, the in-process monitoring and the feed conditions that the step contributes to the commercial control strategy.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8 and the quality risk management principles of ICH Q9, and it supports the drug substance development and manufacture described in ICH Q11 (International Council for Harmonisation 2009, 2023, 2012). The process model, the quality attribute framework and the risk-ranking method are taken from the A-Mab case study (CMC Biotech Working Group 2009).

ICH Q9 treats criticality as a continuum rather than as a binary label. No parameter of this step reaches the critical end of that continuum. Section 9 gives the class of each parameter and the evidence behind it.

1.4 Related documents

Table 1.1 lists the documents this report depends on and the document it rolls up into. Table 1.2 lists the controlled procedures and validated analytical methods cited here.

Table 1.1: Documents related to this report.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-004	Process Characterization Plan (Harvest and Clarification (Step 4))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 1.2: Controlled procedures and validated analytical methods cited in this report. SOP and AMV numbers are placeholders.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Centrifugation followed by depth filtration is the platform recovery operation at Novacyte Biologics and has been used for X-Mab, Y-Mab and Z-Mab as well as for A-Mab clinical supply (CMC Biotech Working Group 2009). The equipment, the media grades and the sequence of operations are common to all four products. The accumulated experience shows that the operation is robust and that it does not alter the product. The A-Mab case study reports the same for its own harvest step (CMC Biotech Working Group 2009). This characterization therefore confirms and bounds the step, and it was not expected to discover a new effect.

A higher centrifugal field raises the sedimentation velocity of cells and debris in proportion to the field, so more of them reach the disk surface within the residence time of the bowl and less passes into the centrate. The shear rate in the feed zone and at the disk stack also rises with bowl speed, and cells damaged by that shear release intracellular protein and DNA into the centrate. The parameter trades clarity against lysis, and the useful operating region is the one where the first effect has saturated and the second has not yet started.

The depth filter has a finite capacity for solids and for the fine debris and colloids that the centrifuge passes. Loading it further raises the pressure drop across the media, lets more fine particulate and turbidity through as the media saturate, and at the extreme forces retained material through the pores. Load per unit area, and not batch volume, is what tracks that accumulation, because two batches of equal volume filtered through different installed areas saturate the media to different extents.

Post-clarification turbidity is not set by the operator. It is a property of the clarified harvest that the centrifugal field and the depth filter load produce, and it is a measure of the fine particulate and colloidal material that the filters passed. It is carried in the parameter list because that material fouls the Protein A column and shortens resin life, so a limit on it is a control on the capture step rather than on this one.

Host cell protein and DNA in the clarified harvest are the reactor's burden plus whatever is released by cells that break during the operation rather than in the reactor. Harvest does not clear host cell protein. A gentle harvest keeps the load close to what left the reactor and a harsh one adds intracellular protein to it. The depth filter media are positively charged and adsorb a fraction of the DNA and debris the centrifuge passes, so some DNA removal across the filtration stage is expected on mechanistic grounds. No clearance credit is taken for it, because the adsorbed fraction is smaller than the qPCR assay can resolve against the spread of the feed between runs.

Aggregate can rise at harvest only if the antibody unfolds at the shear rate of the centrifuge feed zone or at an air-liquid interface created in a foaming transfer. Both exposures last seconds. Any rise in aggregate across the step is therefore expected to be smaller than the SEC method can resolve.

2.2 Quality attributes in scope

The CQA register lists 0 quality attributes as formed or controlled at this step. The attributes in Table 2.1 are the ones the clarified harvest carries into capture and the ones the mechanisms above could change. They are listed here as the attributes the study measured across the step, not as attributes this step sets or clears. Criticality was assigned by the A-Mab Tool #1 assessment, which combines the impact of the attribute on safety and efficacy with the uncertainty in that impact, and attributes scoring high or very high are called critical (CMC Biotech Working Group 2009).

Table 2.1: Quality attributes measured across harvest and clarification, with their drug-substance acceptance criteria and criticality. None of them takes its value at this step.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60

Each of the three is in scope for a different reason. Aggregate is the most critical of the three, and it is in scope because shear in the centrifuge feed zone can unfold the antibody and drive it into soluble aggregate. Host cell protein is the attribute the condition of this step's output bears on most directly, because cells that lyse under that same shear release intracellular protein into the centrate and raise the load the capture column must clear. Residual DNA is of very low criticality at the drug substance and is in scope because the positively charged depth filter media adsorb DNA, and because DNA fragments and nucleosomes bind to the filter matrix and to the capture resin and raise the pressure drop across both.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every parameter of the train by its potential impact on a critical quality attribute and by its potential to interact with other parameters, and it used the resulting score to assign a study type (CMC Biotech Working Group 2009). Parameters with a credible and interacting route to a critical quality attribute were assigned to

a multivariate design. Parameters with a route to process performance, or with a route to a quality attribute that prior knowledge already bounds, were assigned to univariate assessment. Parameters with neither were carried at their platform ranges without new study.

For harvest and clarification the assessment placed all 3 parameters in the second group. The step has no mechanism by which it forms or removes a quality attribute of the antibody, so the severity axis of the assessment is low for every parameter, and no interaction between two parameters can raise a severity that is already low.

3 Materials and methods

3.1 Scale-down model and its qualification

The characterization was executed on a scale-down model of the commercial harvest operation, qualified under SOP-1001. The model comprises a laboratory disk-stack centrifuge of the same disk geometry as the commercial machine and depth filter disks of the same media grades and the same nominal retention ratings as the commercial filter train.

Two scaling rules define the model. The centrifuge is scaled on the ratio of volumetric feed rate to the equivalent settling area of the bowl, which is the quantity that fixes the size of the smallest particle a machine removes at a given throughput. Holding that ratio constant between scales reproduces the clarification the commercial machine achieves, and it is applied across the whole centrifugal field range studied here rather than at the set-point alone. The depth filters are scaled at constant load per unit filter area, which is the parameter under study and is by construction scale-independent. The feed zone geometry and the transfer arrangement were reproduced at the wall shear rate of the commercial machine, because the feed zone is where cells break and release intracellular protein into the centrate, and where the antibody meets the highest shear rate anywhere in the step.

The qualification compared the model against commercial-scale data from the A-Mab engineering campaign on centrate turbidity, clarified harvest turbidity, product recovery and the host cell protein and DNA content of the clarified harvest. The comparison found no difference between scales that the analytical methods could resolve. The model is therefore suitable for establishing the operating ranges of this step, and the conclusions drawn on it are applied directly to the commercial process.

Two bounds on the model apply to every result that follows. The model does not reproduce the residence time of product in the commercial centrifuge bowl or the hold time of the clarified harvest before capture, and neither was varied in this study. The model also uses filter disks rather than the commercial filter modules, so pressure drop at a given load is representative of the media and not of the module housing.

3.2 Operation

The scale-down runs were operated in the commercial sequence. Culture fluid from the production bioreactor was fed continuously to the centrifuge at the scaled feed rate and at the centrifugal field under study. The centrate was collected and passed immediately to the depth filter train at the load per unit area under study. The clarified

harvest was then passed through a sterilising-grade filter and held at the platform temperature until it was loaded onto the capture column.

Inputs that the platform controls to a fixed specification were held at that specification and were not varied. These include the media grade of the depth filters, the sterilising-grade filter rating, the buffer used to flush the filter train before use, and the temperature of the operation. They are identity and quality controlled under SOP-2006 and are not characterization parameters. The feed to the step is the output of the production bioreactor and was taken at the bioreactor set-point condition, so the impurity burden entering harvest was the burden that PCR-003 reports for a set-point culture.

3.3 Analytical methods

Table 1.2 lists the validated methods used. Turbidity was measured by nephelometry under AMV-3015 on the centrifuge feed, the centrate and the clarified harvest, and it is the primary in-process measure of clarification performance. Host cell protein was measured by ELISA under AMV-3012 and residual DNA by quantitative polymerase chain reaction under AMV-3014, both on the centrifuge feed and on the clarified harvest, so that any change across the step would appear as a difference between two samples of the same batch. Size variants were measured by SEC-HPLC under AMV-3011 on the same two samples, which is how the aggregate mechanism was checked. Protein concentration was measured to close the mass balance.

Table 14.1 in Appendix A gives the performance of the two methods whose variance sets the resolution of this study.

The turbidity method has a precision of 4 % relative standard deviation and a limit of quantitation of 0.5 NTU, and it contributes 38.0% of the total variance of a turbidity result. It resolves the differences this study needed to see, since the set-point value of 5 NTU is an order of magnitude above the limit of quantitation. The host cell protein ELISA has a precision of 9.5 % relative standard deviation, which is the coarsest of the methods used. A change in host cell protein across the step smaller than that is not detectable by it.

3.4 Sampling plan

Each run was sampled at three points. A sample of the centrifuge feed was taken from the bioreactor transfer line, a sample of the centrate was taken after the centrifuge and before the depth filters, and a sample of the clarified harvest was taken after the sterilising-grade filter. Turbidity was measured at all three points. Host cell protein, residual DNA and size variants were measured on the feed and on the clarified harvest, which brackets the whole operation.

Sampling at the centrate as well as at the clarified harvest separates the two stages of the operation. A change in turbidity between feed and centrate is the work of

the centrifuge, and a change between centrate and clarified harvest is the work of the depth filters. Without the intermediate sample the two parameters could not be separated, because the centrifuge and the depth filters both remove particulate and both removals appear in the same turbidity result.

The set-point condition was run at the start and at the end of the study, so that any drift in the model or in the assays over the execution period would be detected as a difference between two nominally identical runs.

3.5 Statistical methods

No model was fitted to the data of this study, because the study is univariate and no designed multivariate experiment was performed. The analysis compares each condition against the set-point condition, and it judges a difference by whether it exceeds the analytical variability of the method that measured it. The precisions in Table 14.1 are the reference for that judgement. Where a difference did not exceed the precision of its method, the result is reported as no detectable change and the bound is given as the precision of the method.

This convention is weaker than a fitted model with a lack-of-fit test. A univariate study establishes the range of one parameter with the others held at their set-points. It cannot detect an interaction, where the effect of one parameter depends on the level of another (CMC Biotech Working Group 2009). The consequence for this step is bounded by the risk basis in Section 2.3 and is discussed in Section 11.

Commercial-scale capability for the attributes reported in Section 8 was estimated by Monte Carlo simulation of 2,000 batches of the whole train, with every parameter of every step drawn from within its normal operating range. Capability is reported as a one-sided process capability index against the drug substance acceptance criterion. The simulation varies every step of the train, so its result is a property of the train.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 lists the parameters carried into the study with their set-points, normal operating ranges and characterized ranges. The characterized range is the knowledge space explored in this study and is not itself a proven acceptable range. The proven acceptable ranges are given in Section 7.

Table 4.1: Parameters of harvest and clarification, with set-points, normal operating ranges, characterized ranges, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Centrifugation (rcf)	g	9,000	8000–10000	6000–12000	KPP	univariate
Depth filter load	L/m ²	120	100–140	80–160	KPP	univariate
Post-clarification turbidity	NTU	5	1–10	1–20	GPP	univariate

The characterized ranges were set wider than the normal operating ranges on the same principle the case study applies, which is that a range studied only to the edge of routine control cannot show what happens when routine control is lost (CMC Biotech Working Group 2009). Centrifugal field was studied from 6,000 to 12,000 g against a normal operating range of 8,000 to 10,000 g. Depth filter load was studied from 80 to 160 L/m² against a normal operating range of 100 to 140 L/m². Post-clarification turbidity was studied from 1 to 20 NTU against a normal operating range of 1 to 10 NTU.

The depth filter load range has a direct engineering meaning at commercial scale. For a batch at the commercial working volume of 15,000 L, the set-point load of 120 L/m² sizes the filter train at about 125 m² of media. The top of the characterized range corresponds to about 94 m² and the bottom to about 188 m². The range therefore spans a factor of two in installed filter area. Doubling the area halves the load each square metre carries and leaves the media further from saturation at the end of the batch.

Figure 4.1 shows each range with its normal operating range and set-point. The pattern is the same for all three parameters. The normal operating range sits inside

the characterized range with margin on both sides, and the set-point sits inside the normal operating range.

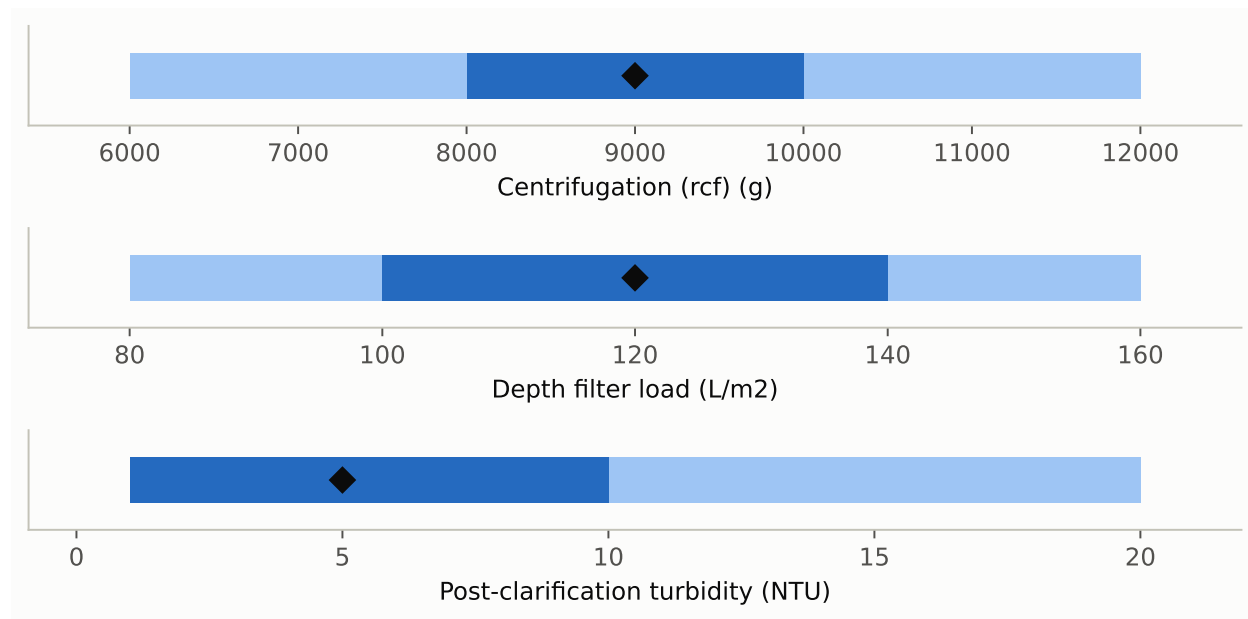


Figure 4.1: Characterized range (light band), normal operating range (dark band) and set-point (marker) for each parameter of harvest and clarification. The normal operating range sits inside the characterized range for all three parameters.

4.2 Univariate characterization design

Each parameter was studied one at a time. The parameter under study was set in turn to the low edge of its characterized range, to its set-point and to the high edge of its characterized range, and every other parameter was held at its set-point. Intermediate settings were added for depth filter load, because the mechanism is a saturation of the media rather than a linear effect and the position of the change matters more than its size.

Post-clarification turbidity was not set at chosen levels, because no valve or drive controls it. It takes the value that the centrifugal field and the depth filter load leave in the clarified harvest. Its range was therefore covered by the settings of those two parameters that produced turbidity across the characterized range, and the range recorded in Table 4.1 is the range of clarified harvest this study produced and assayed.

Every clarified harvest generated in the study was assayed for host cell protein, residual DNA and size variants against the feed of the same run. The comparison is therefore within one batch at every condition, which keeps the variability of the culture out of it.

4.3 Verification at the set-point

The set-point condition was executed at the beginning and at the end of the study. Both executions were assayed on the full panel. These runs serve two purposes. They bound the variability of the step between batches at a fixed condition, which is the basis for the recovery statement in Section 5.1. They also confirm that neither the scale-down model nor the analytical methods drifted over the execution period, since a drift would appear as a difference between two runs at the same condition.

5 Results

5.1 Set-point performance and mass balance

Table 5.1 gives the mass balance and the clarification result at the set-point.

Table 5.1: Product mass balance and clarification result across harvest and clarification at the set-point condition.

Quantity	Value
Product mass into the step	65,596 g
Product mass out of the step	64,071 g
Product mass not recovered	1,525 g
Step yield	97.7%
Cumulative yield through the step	97.7%
Post-clarification turbidity	5 NTU

The step recovered 97.7% of the product mass presented to it. Of the 65,596 g of A-Mab leaving the production bioreactor, 64,071 g reached the capture step and 1,525 g was not recovered. The loss is accounted for by product held up in the sediment discharged from the centrifuge and by product retained in the wetted volume of the depth filter train and its housings. Both losses are volumetric. Installing more filter area to lower the load each square metre carries therefore leaves more product behind in the wetted volume of the train.

Recovery at this step varies between batches with a coefficient of variation of 2.0%, around an expected step yield of 95.0%. The batch reported here recovered above that expected value and within that dispersion. The cumulative yield of the train through this step is 97.7%, and the whole train delivers 83.2%. Figure 5.1 shows where the step sits in the cumulative yield of the train.

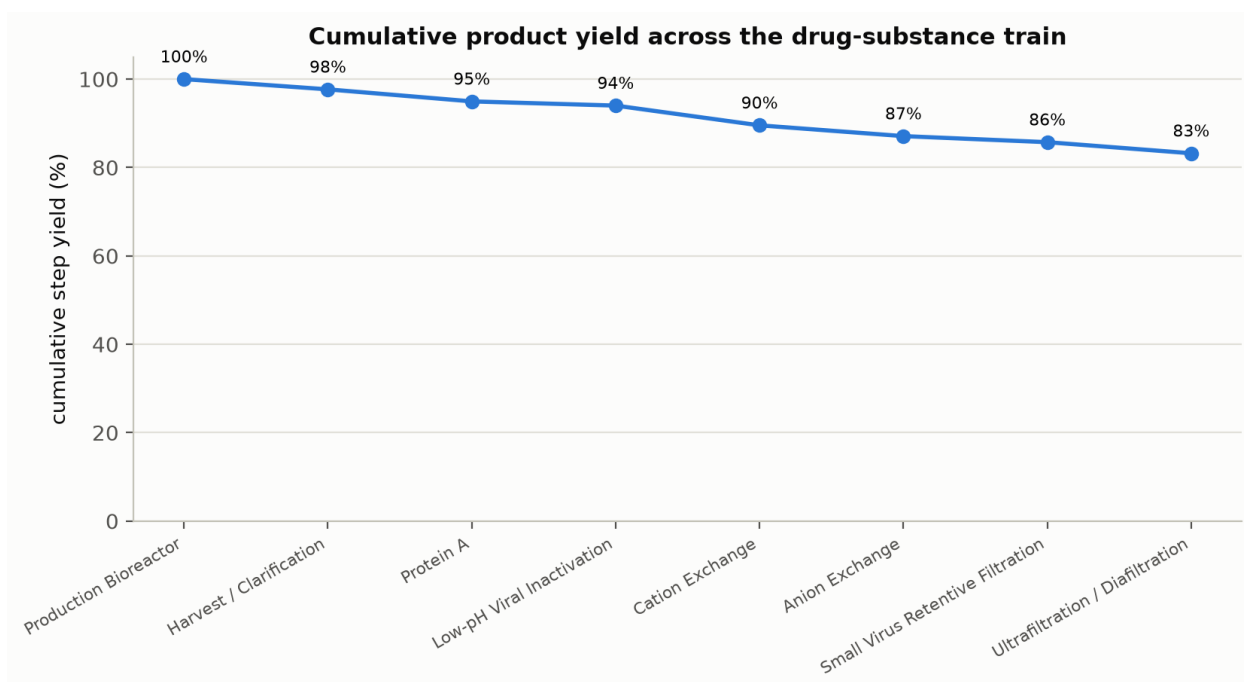


Figure 5.1: Cumulative product yield across the drug-substance train. Harvest and clarification is the first recovery loss after the production bioreactor.

The clarified harvest at the set-point had a turbidity of 5 NTU. That value sits in the middle of the normal operating range of 1 to 10 NTU and well above the limit of quantitation of the method, which is 0.5 NTU. The two set-point executions agreed within the precision of the turbidity method. No drift of the scale-down model or of the assays was detected over the execution period.

5.2 Centrifugal field

Centrifugal field was varied from 6,000 to 12,000 g with feed rate, depth filter load and every other input held at the set-point. Turbidity of the centrate fell as the field rose.

The size of the effect is what decides the range. Across the field range studied, the improvement in centrate turbidity was concentrated at the low end. Above the middle of the range, further increases in field produced no further improvement in the clarified harvest that the turbidity method could resolve. The interpretation is that the population the centrifuge can remove has already been removed by then, and what remains in the centrate is the fine debris and colloidal fraction whose sedimentation velocity is too low for the machine at any field it can reach. That fraction is the depth filter's job, not the centrifuge's.

The lysis mechanism did not appear. Host cell protein and residual DNA in the clarified harvest did not rise with field anywhere in the range studied. For host cell protein the bound is the precision of the ELISA, which is 9.5 % relative standard deviation, so the

result is that any increase in host cell protein caused by shear at the top of the range is smaller than that. Shear damage to a cell population rises steeply once a threshold is passed, and the finding is that 12,000 g is below the threshold for this cell line at this feed rate. The finding does not extend above the range studied.

Centrifugal field is therefore an effective parameter with a saturating benefit and no observed penalty inside its characterized range. The normal operating range of 8,000 to 10,000 g sits above the saturation point of the benefit and below the onset of any detectable lysis.

5.3 Depth filter load

Depth filter load was varied from 80 to 160 L/m² with the centrifugal field at its set-point. The response that moved is post-clarification turbidity, and it moved in the expected direction. Turbidity of the clarified harvest rose with load, and the rise was not linear. Turbidity was flat across the lower part of the range and rose over the upper part, which is the signature of a filter approaching the capacity of its media rather than of a filter passing a constant fraction of what it sees.

Solids accumulate on the media through the batch. As they do, the available media surface falls, and particles that would have been captured early in a batch pass late in the same batch.

The clearest single observation of this effect in the campaign came from a deviation rather than from a planned run. On one run at a depth filter loading near the top of the range studied, post-clarification turbidity reached 12.5 NTU, which is 2.5 NTU above the normal operating range ceiling of 10 NTU and 25 % above it in relative terms. The excursion is far larger than the 4 % precision of the turbidity method, so it is a real excursion and not assay noise. The run reached that value with 7.5 NTU still remaining inside the characterized turbidity range. Section [13.2](#) reports the deviation in full.

Host cell protein, residual DNA and size variants in the clarified harvest did not change with load anywhere in the range studied. The depth filter media adsorb DNA, and a modest reduction in DNA across the filtration stage is expected on mechanistic grounds, but no reduction large enough to resolve against the assay was observed and no clearance is claimed.

The normal operating range of 100 to 140 L/m² sits below the part of the characterized range where turbidity rises. The margin between the top of the normal operating range and the top of the characterized range is the margin that absorbs a filter train sized at nominal capacity running a batch of above-nominal volume.

5.4 Post-clarification turbidity

Post-clarification turbidity across the study ran from 1 to 20 NTU. That range was produced by the settings of the other two parameters and was not imposed.

Turbidity matters to the next step and not to this one. The material it measures is fine particulate and colloidal debris, and that material deposits on the head of the Protein A column. The consequences are a rising column pressure drop over a resin lifetime and a slow loss of dynamic binding capacity. None of these is a quality attribute of A-Mab. Each of them shortens the number of cycles a resin lot runs before its dynamic binding capacity falls below the limit set for the capture step. The limit on post-clarification turbidity is therefore set from resin lifetime and not from a drug substance criterion.

Turbidity is also the only response in this study that moves usefully with either parameter, and it is measured on every batch by a validated method with a precision of 4 % relative standard deviation. It is therefore the natural in-process monitor for the step. A turbidity result above the normal operating range indicates that accumulated solids have begun to let fine particulate through the media. The run described in Section 13.2 is the case this study observed.

5.5 Product quality across the step

No quality attribute of A-Mab changed measurably across harvest and clarification at any condition studied. Host cell protein, residual DNA and size variants were assayed on the centrifuge feed and on the clarified harvest of every run. In every case the difference between the two was within the precision of the method that measured it.

The bounds on that statement are the precisions themselves. For host cell protein the bound is 9.5 % relative standard deviation, which is the coarsest method in the panel, and the limit of quantitation is 1 ng/mg. Any change in host cell protein across the step that is smaller than the first of those numbers would not have been detected. For residual DNA and for aggregate the bounds are the precisions of AMV-3014 and AMV-3011 as established in their validation reports. Changes below these bounds are not excluded.

Two consequences follow. First, the impurity burden presented to Protein A capture is the burden the culture delivered. It rises as culture duration lengthens and rises again as end of culture viability falls, because both leave more lysed cells in the broth, and nothing done at harvest moves it. PCR-003 reports those relationships. Second, this step takes no clearance credit for any impurity and makes no viral clearance claim. The viral clearance claim for A-Mab rests on the low-pH hold, anion exchange and virus filtration, and is consolidated in PCMR-001.

No increase in aggregate was detected between the centrifuge feed and the clarified harvest at any centrifugal field studied. The shear exposure in the feed zone of this operation is brief, and at the fields studied it is not sufficient to unfold the antibody at a rate the SEC method can see.

6 Design space

No design space is defined for harvest and clarification, and none is claimed. A design space is a multivariate region over the parameters that govern a critical quality attribute, within which the attribute is known to remain acceptable (International Council for Harmonisation 2009). No critical quality attribute of A-Mab responds to a parameter of this step, so there is nothing for such a region to be defined against, and the multivariate studies that would define one were not performed.

What bounds this step instead is a set of univariate proven acceptable ranges, one per parameter, supported by the data in Section 5 and reported in Section 7. This is the distinction PDA Technical Report 60 draws when it allows univariate approaches to establish a proven acceptable range while reserving multivariate studies for the interactions between parameters. The A-Mab case study excludes its seed expansion steps from the design space on the same ground (CMC Biotech Working Group 2009).

Movement inside the ranges in Table 7.1 is movement inside a characterized region and is managed under the site change management system (International Council for Harmonisation 2008). Movement outside them requires assessment, and because no multivariate model of this step exists, that assessment cannot be made by prediction. It requires data.

7 Proven acceptable ranges

Table 7.1 gives the proven acceptable range of each parameter with the basis on which it is proven.

Table 7.1: Proven acceptable ranges for the parameters of harvest and clarification, with the basis on which each range is proven.

Parameter	Unit	Set-point	NOR	Proven acceptable range	Acceptance basis
Centrifugation (rcf)	g	9,000	8000–10000	6000–12000	Clarification performance and product recovery
Depth filter load	L/m ²	120	100–140	80–160	Post-clarification turbidity and product recovery
Post-clarification turbidity	NTU	5	1–10	1–20	Condition of the feed to Protein A capture

The proven acceptable range of each parameter is the range studied, because every condition studied produced a clarified harvest that met the requirements of the next step and no condition studied changed a quality attribute of the antibody. The value in the turbidity row is the turbidity the step delivers when the other two parameters are at their set-points. Its range is an acceptance range on the clarified harvest, and no operator moves it.

Three bounds apply to these ranges. Each was established with the other parameters held at their set-points, so none of them is a statement about the corners where two parameters are at their edges at once. Each rests on a scale-down model qualified against the attributes listed in Section 3.1, and the model bounds recorded there apply. Each is a range over which the step performs. None of them is a range over which a critical quality attribute has been shown to remain in limit, because no critical quality attribute of A-Mab responded to any parameter of this step.

The normal operating ranges in Table 7.1 are narrower than the proven acceptable ranges in every case, and they are unchanged from the ranges used for clinical manufacture. The margin between the two is what absorbs normal equipment variability and measurement uncertainty at commercial scale without leaving the characterized region.

8 Process capability and robustness

Process capability is a property of a quality attribute and not of a process parameter. No quality attribute of A-Mab takes its value at this step, so no capability is attributed to harvest and clarification. Table 8.1 gives the commercial-scale capability of the three attributes the clarified harvest carries into capture. Each of those values is produced by the steps that clear the attribute.

Table 8.1: Commercial-scale capability of the attributes carried forward from harvest, estimated by Monte Carlo simulation of the whole train with every parameter inside its normal operating range. The capability is delivered by the downstream steps that clear these attributes.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Aggregates (HMW)	H	upper	0-5	0.753 ± 0.088	16.1
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0-0.001	0.0001 ± 0	10.8

Of the three attributes, Host Cell Protein (HCP) has the tightest capability at a process capability index of 6.14, and it is cleared by Protein A capture, cation exchange and anion exchange as reported in PCR-005, PCR-007 and PCR-008. The tightest capability anywhere in the drug substance is 1.51 and it belongs to a viral clearance attribute set at another step, which PCMR-001 consolidates.

The robustness of this step is its own recovery and its own clarification, and both were stable. Recovery varied between batches with a coefficient of variation of 2.0% at a fixed condition. Clarification performance was unchanged across the middle and upper part of the centrifugal field range and across the lower part of the depth filter load range. The one condition at which performance degraded was high depth filter loading, and what degraded was the turbidity of the clarified harvest. No attribute of the antibody changed with it.

9 Parameter classification

Parameters were classified under SOP-4001 using the decision logic of the A-Mab case study (CMC Biotech Working Group 2009). A parameter whose variability affects a critical quality attribute is a critical process parameter or a well-controlled critical process parameter, according to the risk that it leaves its acceptable range. A parameter that affects process consistency or performance without affecting a quality attribute is a key process parameter. A parameter with neither effect is a general process parameter.

- Centrifugation (rcf) was classified as a key process parameter. A higher field removes more cells and debris and therefore leaves a smaller solids load for the depth filters to hold. No effect on any quality attribute was detected across 6,000 to 12,000 g, and in particular no shear-driven release of host cell protein or DNA was detected at the top of that range. The parameter is set on the machine and held by its drive control, so the risk of leaving the range is low.
- Depth filter load was classified as a key process parameter. Turbidity of the clarified harvest rises as loading rises, and the particulate that turbidity measures deposits on the head of the capture column and shortens the number of cycles the resin lot runs. No effect on any quality attribute was detected across 80 to 160 L/m². Loading is fixed by the installed filter area and the batch volume, both of which are known before the batch starts, so the parameter is controlled by design rather than in real time.
- Post-clarification turbidity was classified as a general process parameter. It is an output of the step and not a setting, no quality attribute of A-Mab changes with it, and the particulate it measures deposits on the capture column, where the resin cycling limit bounds the consequence. It is measured on every batch under AMV-3015, where a result above the normal operating range identifies a batch whose depth filter media were loaded past the point at which they hold fine particulate.

Of the 3 parameters, 2 are key process parameters and 1 is a general process parameter. No parameter of this step is a critical process parameter or a well-controlled critical process parameter. No quality attribute of the antibody is formed or cleared at this step, and none responded to a parameter of it. Key process parameters are included in the control strategy even though they do not affect product quality, because controlling them keeps the process consistent and predictable (CMC Biotech Working Group 2009).

10 Contribution to the control strategy

The step contributes four elements to the commercial control strategy.

The first is the two key process parameter ranges. Centrifugal field is controlled to 8,000 to 10,000 g and depth filter load to 100 to 140 L/m², both inside the ranges proven in Section 7. Depth filter load is controlled by sizing the installed filter area for the batch volume, which at the set-point loading is about 125 m² for a batch at the commercial working volume.

The second is in-process monitoring. Post-clarification turbidity is measured on every batch under AMV-3015 and is the in-process acceptance criterion the clarified harvest meets before it is loaded onto Protein A capture. A result above the normal operating range sends the batch to an assessment of its depth filter loading, because loading the media past their retention capacity is what raised turbidity in the one run where turbidity rose.

The third is a life-cycle control on the equipment and the consumables. Depth filter media lots are qualified before use by a pre-use flush with a turbidity acceptance criterion, which is the control that detected DEV-004-01. Resin cycling limits at the capture step account for the particulate the clarified harvest carries onto the column, and PCR-005 reports the cycling limit.

The fourth is a specification on the load presented to the next step. Protein A capture receives a clarified harvest with a defined turbidity range and a host cell protein and DNA burden that rises with culture duration and with the fraction of cells that lysed before harvest. PCR-005 characterizes the capture step on that load and not on an idealized one.

No release test of the drug substance depends on this step alone. The attributes in Table 2.1 are tested at the drug substance and are controlled by the steps that clear them. The contribution of this step is consolidated in PCMR-001.

11 Discussion

The study confirms what prior knowledge predicted and bounds it with data. Harvest and clarification recovers 97.7% of the product presented to it, delivers a clarified harvest at 5 NTU at the set-point, and changes no quality attribute of A-Mab within the ranges studied. What the step changes is the particulate and impurity load that the capture column receives.

The mechanistic picture that emerges is a step operated comfortably inside two saturating effects. The centrifuge is operated above the field at which its benefit saturates and below the field at which shear damage to the cell population would begin, and neither edge was reached inside the characterized range. The depth filter train is operated below the load at which the media begin to saturate, and that edge was reached inside the characterized range and was observed. One edge lies inside the studied region and one lies outside it, and the two statements therefore do not carry the same confidence. The depth filter load limit is supported by an observed transition. The centrifugal field limit is supported by the absence of a transition, and that absence is bounded by the precision of the assays that looked for it.

Confidence in the scale-down model rests on the two scaling rules in [Section 3.1](#) and on the comparison against commercial-scale data. The residual risk is the shear rate in the feed zone, which a laboratory centrifuge reproduces less faithfully than it reproduces clarification, and which is what would break cells and unfold antibody if anything did. The study looked for the consequence of shear directly, on every run, and did not find it.

Three limitations bound this report. The study is univariate, so it says nothing about the corner where a low centrifugal field and a high depth filter load meet. That corner is the one the mechanism predicts to be worst, because a poorly clarified concentrate arriving at an already heavily loaded filter train raises turbidity by both routes at once. The risk it carries is a process performance risk and not a quality risk, and it is managed by controlling both parameters independently within their normal operating ranges. The second limitation is that the conclusions on host cell protein, DNA and aggregate are bounded by assay precision and not by zero. The third is that hold time of the clarified harvest before capture was not varied, so this report does not bound it, and it is controlled at the platform limit under SOP-2006.

Two deviations were recorded and both were retained. Neither affected product quality and neither altered a conclusion of this report. One of them supplied the study's clearest observation of the depth filter loading mechanism, reported in [Section 5.3](#) with the other observations and again in [Section 13](#) as a deviation.

12 Conclusions

Harvest and clarification was characterized on a qualified scale-down model over the ranges in Table 4.1, and the step performs as the platform experience predicted.

The step recovered 97.7% of the product mass presented to it and delivered a clarified harvest at 5 NTU at the set-point. No quality attribute of A-Mab was formed, modified or cleared across the step at any condition studied, and no clearance credit and no viral clearance claim is taken for it. Proven acceptable ranges were established for all 3 parameters and are given in Table 7.1. No design space is defined for this step, because no critical quality attribute of A-Mab responded to any parameter of it.

Of the 3 parameters, 2 were classified as key process parameters and 1 as a general process parameter. The step contributes two parameter ranges, in-process turbidity monitoring, depth filter media lot qualification and a defined feed condition for Protein A capture to the commercial control strategy. During execution 2 deviations were recorded, investigated and dispositioned as retained. The results of this report are consolidated in PCMR-001.

13 Deviations from the plan

During execution 2 deviations from PCP-004 were recorded. Table 13.1 lists them. Both were dispositioned as retained, which means that the affected data were used as reported and that no run was repeated.

Table 13.1: Deviations recorded during execution of PCP-004.

Devia- tion	Summary	Detected during	Disposi- tion
DEV- 004-01	Depth-filter lot pre-use flush turbidity above alert	pre-use filter flush	retained
DEV- 004-02	Post-clarification turbidity excursion above NOR on one run	in-process turbidity monitoring	retained

13.1 DEV-004-01 — Depth-filter lot pre-use flush turbidity above alert

The pre-use flush of depth filter media lot LOT-DF-4408 gave a flush turbidity of 6 NTU, above the alert level set for the flush in SOP-2006. The lot was within its expiry date of 2027-01-31 and had been released by the vendor against its own certificate. The deviation was detected at the flush itself, before any culture fluid contacted the media.

The investigation attributed the result to an insufficient initial flush volume for this media grade. Depth filter media shed a small quantity of fines when first wetted, and the flush volume specified in SOP-2006 was found to be adequate for the grades in routine use but marginal for this one.

The measurement is a pre-use measurement of the filter train, taken before product was present, so no product was exposed to the fines that the reading reflects. Even at the flush value observed, a clarified harvest at 6 NTU would have sat inside the normal operating range for post-clarification turbidity, whose ceiling is 10 NTU. The lot was used and the data from the affected run were retained. The investigation recommended that the minimum pre-use flush volume in SOP-2006 should be stated per media grade rather than as a single value for the filter train.

13.2 DEV-004-02 — Post-clarification turbidity excursion above the normal operating range

On one run at a depth filter loading near the top of the range studied, post-clarification turbidity reached 12.5 NTU. The normal operating range for that attribute has a ceiling of 10 NTU, so the excursion was 2.5 NTU or 25 % above it. The turbidity method has a precision of 4 % relative standard deviation, so the excursion is about 6 times the analytical variability of the result and is real. It was detected by the in-process turbidity monitor under AMV-3015.

The investigation attributed the excursion to depth filter loading near the maximum of the range under study, by the mechanism described in [Section 5.3](#).

What rose was the turbidity of the clarified harvest, which measures the particulate carried to capture. No attribute of the antibody rose with it. The host cell protein, residual DNA and size-variant results for that run were within the precision of their methods against the feed of the same run, and the turbidity itself stayed inside the characterized range, which extends to 20 NTU. The extra particulate the run carried onto the capture column consumes part of the cycle life of the resin, and PCR-005 sets the cycling limit that bounds it.

The run and its data were retained. The ceiling of the normal operating range for depth filter load is set at 140 L/m², below the loading at which that run was executed.

14 Appendix A — Analytical methods summary

No design matrix is reproduced, because no designed multivariate experiment was performed for this step. Table 14.1 gives the performance of the validated methods whose variance bounds the conclusions of this report. The remaining methods in Table 1.2 are covered by their own validation reports.

Table 14.1: Analytical method performance for the methods that bound the conclusions of this report.

Refer- ence	Method	Precision (% RSD)	LOQ	Accuracy (%)	Analytical share of total variance
AMV- 3015	Turbidity by Nephelometry (NTU)	4	0.5 NTU	98	0.38
AMV- 3012	Host-Cell Protein by ELISA	9.5	1 ng/mg	95	0.4

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